

Unique factorization beyond $\mathbb{Z}$.

Last time!

Let D be an integral domain.

Defn (Divisibility) For $\alpha, \beta \in D$ with $\alpha \neq 0$, we say α

divides β and write $\alpha \mid \beta$ if there is $r \in D$ s.t. $\beta = \alpha r$.

Defn (unit) A unit $u \in D$ is a nonzero element s.t. there

is $v \in D$ s.t. $uv = vu = 1$.

Defn (Associate) An associate of α is of the form $u\alpha$ with

a unit $u \in D$.

Properties of divisibility

1. Reflexivity: $\alpha \mid \alpha$

2. Anti-symmetry: $\alpha \mid \beta, \beta \mid \alpha \Rightarrow \beta$ is an associate of α .

3. Transitivity: $\alpha \mid \beta, \beta \mid \gamma \Rightarrow \alpha \mid \gamma$

4. Linearity: $\alpha \mid \beta, \alpha \mid \gamma \Rightarrow \alpha \mid x\beta + y\gamma$ for any $x, y \in D$

5. Size: $\alpha \mid \beta$ ($\beta \neq 0$) $\Rightarrow \alpha$ is "smaller" than β ?

Recall the proof of FTA:

Size $\Rightarrow$ existence

$DA \Rightarrow \mathbb{Z}$ is a PID $\stackrel{\text{gcd}}{\Rightarrow}$ Bezout's lemma

$\Rightarrow$ Euclid's lemma $\Rightarrow$ uniqueness

Arithmetic in $\mathbb{Z}[\sqrt{d}]$

Let $d \in \mathbb{Z}$. Define

$$\mathbb{Z}[\sqrt{d}] = \{x + y\sqrt{d} : x, y \in \mathbb{Z}\}.$$

When $d = -1$, a complex number $x + y\sqrt{-1} = x + yi$ with

$x, y \in \mathbb{Z}$ is called a Gaussian integer. The set of all Gaussian

integers is denoted by

$$\mathbb{Z}[i] = \{x + yi : x, y \in \mathbb{Z}\}.$$

Note: $\mathbb{Z}[\sqrt{d}]$ is an integral domain under $+$ and $\times$:

$$(x_1 + y_1\sqrt{d}) + (x_2 + y_2\sqrt{d}) = (x_1 + x_2) + (y_1 + y_2)\sqrt{d}.$$

$$(x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d}) = (x_1x_2 + y_1y_2d) + (x_1y_2 + x_2y_1)\sqrt{d}.$$

Defn (Norm) For $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$, the norm of α is defined as

$$N(\alpha) = (x + y\sqrt{d})(x - y\sqrt{d}) = x^2 - dy^2 \in \mathbb{Z}.$$

Defn (Conjugate) The conjugate of $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ is

$$\bar{\alpha} = x - y\sqrt{d}.$$

$$\text{So } N(\alpha) = N(\bar{\alpha}) = \alpha\bar{\alpha}.$$

From now on, we will assume that d is not a square,

so that $\mathbb{Z}[\sqrt{d}]$ strictly contains $\mathbb{Z}$.

Observation: If d is not a square, then

$$N(\alpha) = 0 \Leftrightarrow \alpha = 0.$$

$$\alpha = x + y\sqrt{d} = 0 \Rightarrow x = -y\sqrt{d}$$

$$\Rightarrow x^2 = y^2 d$$

$$\Rightarrow N(\alpha) = x^2 - y^2 d = 0.$$

$$N(\alpha) = 0 \Rightarrow x^2 - y^2 d = 0 \Rightarrow x^2 = y^2 d.$$

① $|d| = 1$: $d = -1$. But $x^2 = -y^2$. So $x = y = 0$.

② $|d| > 1$: If $y = 0$, $x = 0$ ✓

Assume $y \neq 0$, so that $x \neq 0$. If $d < 0$, then

$x^2 = y^2 d$ implies $x = y = 0$, a contradiction.

If $d > 0$, then there is a prime $p|d$ s.t. $e_p(d)$ is odd.

But this is impossible, because $e_p(x^2)$ and $e_p(y^2)$ are even:

$$x^2 = y^2 d \text{ implies } \underbrace{e_p(x^2)}_{\text{even}} = \underbrace{e_p(y^2)}_{\text{even}} + \underbrace{e_p(d)}_{\text{odd}}.$$

□

Lem (Multiplicativity of conjugates) For any $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$,

$$\overline{\alpha\beta} = \overline{\alpha} \overline{\beta}.$$

Pf. Assume $\alpha = x_1 + y_1\sqrt{d}$, $\beta = x_2 + y_2\sqrt{d}$.

$$\alpha\beta = (x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d}) = (x_1x_2 + y_1y_2d) + (x_1y_2 + x_2y_1)\sqrt{d}.$$

$$\Rightarrow \overline{\alpha\beta} = (x_1x_2 + y_1y_2d) - (x_1y_2 + x_2y_1)\sqrt{d}.$$

$$\text{Also, } \overline{\alpha} \overline{\beta} = (x_1 - y_1\sqrt{d})(x_2 - y_2\sqrt{d})$$

$$= (x_1x_2 + y_1y_2d) + (-x_1y_2 - x_2y_1)\sqrt{d} = \overline{\alpha\beta}.$$

□

Prop (Multiplicativity of norm). For any $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$,

$$N(\alpha\beta) = N(\alpha)N(\beta).$$

Consequently, if α/β with $\beta \neq 0$, then $|N(\alpha)| \leq |N(\beta)|$.

pf. $N(\alpha\beta) = \alpha\beta \overline{\alpha\beta} = \alpha\beta \overline{\alpha} \overline{\beta} = (\alpha\overline{\alpha})(\beta\overline{\beta}) = N(\alpha)N(\beta).$

(Diagram: A red arrow points from $\overline{\alpha\beta}$ to $\overline{\alpha}\overline{\beta}$ with the label "lem" below it. A blue arrow points from $\overline{\alpha}\overline{\beta}$ to $\overline{\alpha}\overline{\beta}$.)

If α/β with $\beta \neq 0$, then there is some $\gamma \in \mathbb{Z}[\sqrt{d}]$ s.t.

$$\alpha = \beta\gamma. \text{ So } N(\alpha) = N(\beta\gamma) = N(\beta)N(\gamma). \text{ Thus } N(\beta) \mid N(\alpha).$$

$$\text{But } \beta \neq 0 \Rightarrow N(\beta) \neq 0. \text{ So } |N(\alpha)| \leq |N(\beta)|.$$

□

Cor (units) $\alpha \in \mathbb{Z}[\sqrt{d}]$ is unit if and only if $|N(\alpha)| = 1$.

Pf. If α is a unit, there is some $\beta \in \mathbb{Z}[\sqrt{d}]$ s.t. $\alpha\beta = 1$.

Taking norms: $N(\alpha)N(\beta) = N(1) = 1$. So $N(\alpha) \mid 1$. Thus

$$|N(\alpha)| = 1.$$

Conversely, if $N(\alpha) = \varepsilon$, where $\varepsilon = \pm 1$, then $\alpha\bar{\alpha} = \varepsilon$.

So $\alpha(\underbrace{\varepsilon\bar{\alpha}}_{\beta}) = \varepsilon^2 = 1$. By definition, α is a unit. □

Rmk. Let $\alpha \in \mathbb{Z}[\sqrt{d}]$. Then β is an associate of α if and only

if $\beta \mid \alpha$ and $|N(\beta)| = |N(\alpha)|$.

Pf. Exercise.

Cor (primes) $\alpha \in \mathbb{Z}[\sqrt{d}]$ is prime if $|N(\alpha)|$ is a prime number.

pf. Let $\beta | \alpha$. Note that $\alpha \neq 0$ because $|N(\alpha)| > 0$.

By multiplicativity of norm, $N(\beta) | N(\alpha)$, so

① $|N(\beta)| = 1$: β is a unit.

② $|N(\beta)| = |N(\alpha)|$: β is an associate of α .

Hence α is prime. □

Warning! It is not true that a prime $\alpha \in \mathbb{Z}[\sqrt{d}]$ has

a prime norm $|N(\alpha)|$. Think about $2, 3, 1 \pm \sqrt{-5} \in \mathbb{Z}[\sqrt{-5}]$.

Try to show that they are prime but have composite norms

in absolute value.